

The background of the slide is a deep space image. It features a prominent, thick, reddish-pink filament or structure stretching diagonally from the top left towards the right. Several bright, yellowish-white stars are scattered across the dark field. In the lower right, there is a faint, blueish-white spiral galaxy. The overall tone is dark and cosmic.

OrbitTails

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Astro 330 Final Project

Science Background: Ram Pressure Stripping

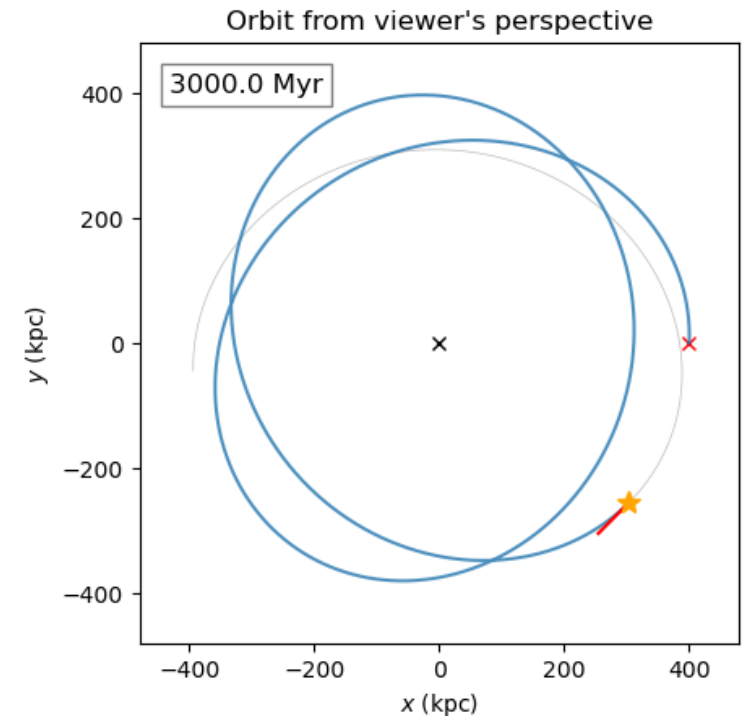
- Gas removal mechanism in galaxy clusters and driver of galaxy evolution
- Creates extended gas tails, measurable in H α
- Tail angle: key parameter for understanding stripping stage and orbital motion

Measurements of tail angle are complicated by our observing angle!



OrbitTails: A User-Friendly Interface for Calculating and Comparing Tail Angles

- Generate cluster gravitational potential and simulate the orbit of a galaxy in that potential.
- Compute 3D tail directions, tangential to galaxy orbit.
 - Tail angle: measured between the tail and the radial direction toward cluster center.
- Project the orbit and tail directions into the viewer's 2D perspective and recompute tail angles.
- Plot orbits and compare 2D and 3D tail angles.



Basic Structure of the Codebase

- Orbit.py: main simulation and angle calculations
 - Orbit
 - Projected_Orbit
- Web_app.py: script for streamlit interface
- Transforms.py: helper functions for coordinate and angle transforms
- Plotting_utils.py: helper functions for plotting orbits and tails
- Tutorial.ipynb
- README.md

Demo!
Visit
orbittails.streamlit.app

